

The Quality Ceiling Is the Task, Not the Model Size

A Reproducible CPU/GPU Benchmark of Search-Relevance Judges

Ferosh Jacob & Jiho Noh

Department of Computer Science, Kennesaw State University

Presenter / PI: Ferosh Jacob

Contact: fjacob1@kennesaw.edu · jnoh3@kennesaw.edu

Metric: Quadratic Weighted Kappa (QWK), 95% CI

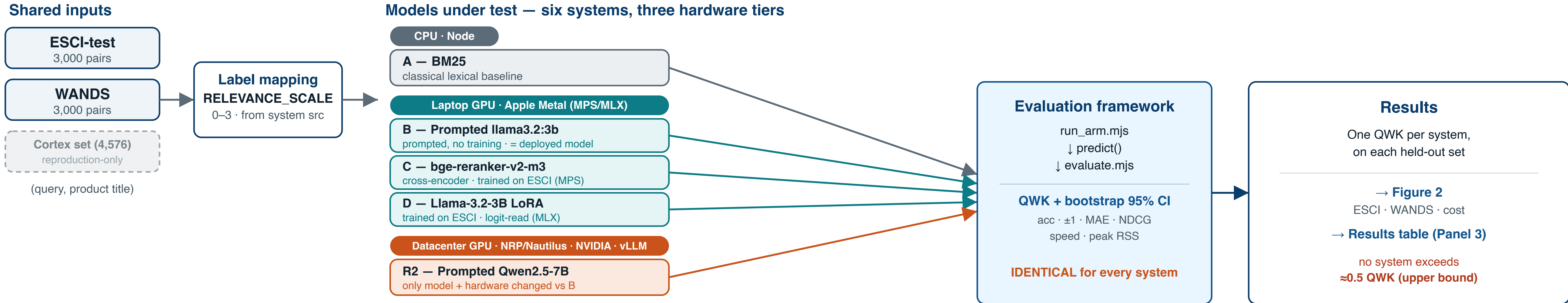


Figure 1. One evaluation framework, six systems, three hardware tiers. Shared inputs are label-mapped, then each **model under test** (CPU; laptop GPU on Apple Metal; datacenter GPU on NRP/Nautilus · vLLM) produces predictions that flow into a **single evaluation framework whose code is byte-identical for every system**, so any QWK movement is attributable to the model & hardware, not pipeline drift. Results converge on Figure 2 and the results table (Panel 3). Vector diagram, built for print.

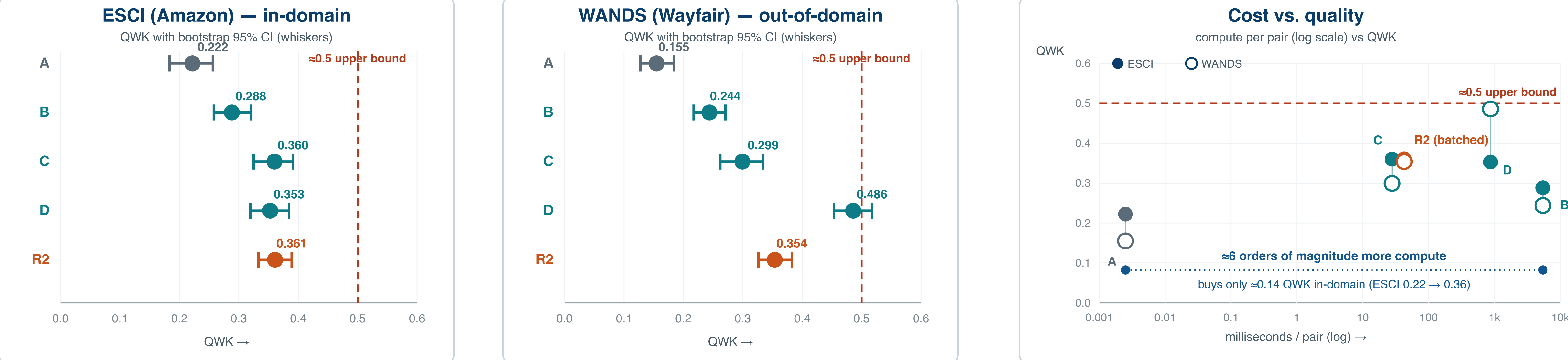


Figure 2. QWK per system with bootstrap 95% CI whiskers on each independent held-out set (**ESCI** in-domain; **WANDS** out-of-domain), and **cost vs. quality** on a log compute axis (filled = ESCI, open = WANDS). The dashed line marks an **empirical upper bound** (≈ 0.5 QWK) that no system exceeds — not a claim that every system reaches it; a true task ceiling would need human-human agreement (future work). Across ≈ 6 orders of magnitude in per-pair compute, in-domain QWK moves only ≈ 0.14 . R2 is datacenter *batched* throughput, a different latency regime (see L6). Every plotted value is the exact figure from the results table (Panel 3).

0 Abstract

E-commerce search must grade how well each product matches a query on an ordinal **0–3** scale (0 Irrelevant · 1 Marginal · 2 Relevant · 3 Perfect). We ask an honest, reproducible question: **how good a relevance judge can run entirely on a laptop, and when is a GPU actually worth it?**

Using one shared evaluation framework and identical held-out sets (Amazon **ESCI** and Wayfair **WANDS**, 3,000 pairs each, independent human labels), we benchmark six systems across three hardware tiers: a **CPU** lexical baseline (BM25); three **laptop-GPU** systems on Apple Metal (a prompted `llama3.2:3b`, a fine-tuned `bge-reranker-v2-m3` cross-encoder, and a `Llama-3.2-3B` LoRA read via label-token logits); and a **datacenter-GPU** system (`Qwen2.5-7B-Instruct` prompted via vLLM on an NRP/Nautilus GPU).

The headline metric is **Quadratic Weighted Kappa (QWK)** with bootstrap 95% CIs. **Key finding:** every system lands in a narrow **~ 0.3 – 0.5 QWK band**. Scaling a prompted judge 3B $\rightarrow$ 7B on a real GPU lifted QWK (+0.073 ESCI, +0.110 WANDS) enough to *match* the fine-tuned cross-encoder in-domain with **no training** — but did **not** break out, and the best out-of-domain result (the 3B LoRA, WANDS 0.486) still stands. **The ceiling is the task, not the model size**. We report every loss beside every win.

1 Background & Problem

- The task.** A production relevance system (codename *Cortex*) grades each (query, product title) pair on the 0–3 RELEVANCE_SCALE. Its deployed model is an LLM-as-judge, one model call per pair.
- The question.** Pointwise relevance grading is an **ordinal classification** problem. How well can it be done on a laptop, and where does reaching for the GPU — first the laptop's integrated GPU, then a datacenter card — actually pay off in accuracy?
- Why it's hard / matters.** Relevance is **intent, not word overlap**; a “2 vs 1” boundary is genuinely subjective even for humans. An on-device judge is cheaper and private, but must be shown to **beat the local judge it would replace**, on independent labels — not merely reproduce it.

2 Methods

2.1 One shared evaluation framework

Every system runs the identical pipeline so the six are directly comparable. **Inputs:** (query, product title) only — the exact signal the production judge uses. **Label mapping** imports RELEVANCE_SCALE from the system's own source (never re-typed): ESCI E $\rightarrow$ 3, S $\rightarrow$ 2, C $\rightarrow$ 1, I $\rightarrow$ 0; WANDS Exact $\rightarrow$ 3, Partial $\rightarrow$ 2, Irrelevant $\rightarrow$ 0 (grade 1 deliberately unpopulated).

Metric (headline): QWK — ordinal agreement that penalizes a 3-vs-0 miss far more than a 3-vs-2 miss — with a **1,000-resample bootstrap 95% CI**. Secondary: exact accuracy, ± 1 accuracy, MAE, NDCG. Speed and peak RSS measured per system.

2.2 Datasets & avoiding circular evaluation

ESCI (Amazon; native query-disjoint split, verified 0 query overlap) and **WANDS** (Wayfair furniture; eval-only) carry the accuracy verdict — independent human-derived labels. We also keep the system's 4,576 past validations, but use them *only* as a reproduction check, never for accuracy: **92.3%** of those grades were themselves produced by the local `llama3.2:3b`, and here they are graded by a *different* model, so “agreement” there measures whether a model reproduces the local model's pseudo-labels, not whether it is correct. This is a **label-provenance** issue (one model grading another model's labels) — **not** the same as IR “pool bias” or test-collection reusability, where judgments are pooled across systems. Every accuracy claim here is on ESCI/WANDS human labels only.

2.3 The six systems, by hardware tier

Tier	System & approach
CPU	A — BM25 — classical lexical baseline. CPU, ≈ 0.003 ms/pair.
Laptop GPU	B — Prompted <code>llama3.2:3b</code> — LLM-as-judge via Ollama = deployed on-device model. Metal, $\approx 5,440$ ms/pair.
Laptop GPU	C — <code>bge-reranker-v2-m3</code> — cross-encoder fine-tuned on ESCI, 4-class argmax (PyTorch-MPS). ≈ 28 ms/pair.
Laptop GPU	D — <code>Llama-3.2-3B LoRA</code> — LoRA fine-tune, read label-token logits (no free generation, Apple MLX). ≈ 360 – 873 ms/pair.
Datacenter GPU	R2 — Prompted <code>Qwen2.5-7B-Instruct</code> — SAME prompt+parser+QWK as System B, bigger model, via vLLM. NVIDIA GPU, batched.

Systems A–D (Round 1) run on an Apple M-series laptop (16 GB); the “GPU” there is the integrated Apple GPU (Metal) — no hand-written CUDA/Metal kernels. **Round-2 (R2)** is the controlled scale-up: it changes **only the model and the hardware**, holding prompt, parser, QWK code and eval sets **byte-for-byte identical** to System B (system-prompt sha256 matched on laptop and in-cluster). Preferred model was **Llama-3.1-8B-Instruct** (clean same-family 3B $\rightarrow$ 8B), but it is gated with no HF_TOKEN, so open **Qwen2.5-7B-Instruct** was used (stated confound: family and size change). Infra: 1 $\times$ NVIDIA GPU on NRP/Nautilus via vLLM chat, namespace kennesaw-state-fjacob, RTX-3090/A10-class node; model load 103.8 s. Parse failures are excluded from QWK, **never coerced** to 0.

Key finding

Two fine-tunes and a 2.3 $\times$ -bigger prompted judge all cluster in the **~ 0.3 – 0.5 QWK band**. Scaling helped *within* the band but did not break *out* of it — the single best out-of-domain number, the 3B LoRA's **WANDS 0.486**, still stands. **The ceiling is the TASK, not the model size**.

3 Results table — QWK on the independent held-out sets (95% CI)

System	ESCI QWK (95% CI)	WANDS QWK (95% CI)	ESCI acc	WANDS acc
A BM25	0.222 0.183–0.256	0.155 0.128–0.184	0.401	0.377
B Prompted <code>llama3.2:3b</code> (deployed)	0.288 0.258–0.320	0.244 0.217–0.271	0.317	0.310
C bge cross-encoder (fine-tuned)	0.360 0.325–0.391	0.299 0.262–0.334	0.517	0.473
D Llama-3.2-3B LoRA (logit-read)	0.353 0.320–0.385	0.486 0.453–0.517	0.483	0.603
R2 Prompted <code>Qwen2.5-7B</code> (GPU)	0.361 0.333–0.389	0.354 0.326–0.382	0.343	0.332

Exact QWK: A 0.22204 / 0.15478 · B 0.28823 / 0.24383 · C 0.35996 / 0.29928 · D 0.35275 / 0.48581 · R2 0.36095 / 0.35367.

Test sets: 3,000 pairs each. System B parsed 2,993 (ESCI) / 2,989 (WANDS) — 7 / 11 parse failures excluded, never coerced. R2 parsed 2,997 (ESCI) / 3,000 (WANDS) — 3 / 0 failures. Highlighted cells mark the top in-domain (C, ESCI 0.360), the best out-of-domain (D, WANDS 0.486), and the datacenter-GPU in-domain tie (R2, ESCI 0.361). Compare systems **WITHIN** a dataset only (limitation L2).

4 Results — Round-2 read

Scaling the prompted judge 3B $\rightarrow$ 7B on a datacenter GPU lifted QWK — but stayed in the band. vs System B (same method, 3B): **+0.073 on ESCI** (0.288 $\rightarrow$ 0.361) and **+0.110 on WANDS** (0.244 $\rightarrow$ 0.354). With **no fine-tuning**, the prompted 7B:

- ties** the fine-tuned cross-encoder on ESCI (0.361 vs 0.360), and
- beats** it on WANDS out-of-domain (0.354 vs 0.299).

But it did NOT break the ceiling. The 7B lands at ~ 0.35 on both sets — squarely inside the same ~ 0.3 – 0.5 band every Round-1 system occupied. A 14B/70B prompted judge was **deliberately not run**: the trend showed diminishing returns inside the band, so more GPU was not justified.

4.3 Speed & memory — why the laptop still wins deployment

System	Latency (ms/pair)	Peak mem
A BM25	≈ 0.003 lexical, CPU	~ 0.11 GB
B Prompted 3b	$\approx 5,440$ / 5,200 conc-4; ~ 0.73 – 0.77 pairs/s	Ollama ~ 5 – 6 GB
C bge	27.9 / 29.6 p95 ~ 38 – 40 ms; 0 parse fail	~ 1.06 GB
D LoRA	873 / 360 p95 3,220 / 1,430 ms	~ 2.67 GB
R2 Qwen 7B	≈ 43 / 30 batched; 103.8 s load	≥ 24 GB GPU

Deployment winner: bge (System C) — matches/beats the on-device judge on accuracy at **~ 1 – 2 orders of magnitude faster** (~ 28 ms vs thousands of ms/pair) in **$\sim 5\times$ less memory**, never malfunctioning output. **Out-of-domain winner:** the LoRA (System D) — WANDS 0.486 (a 120-pair re-run reproduced ~ 0.448); cost ~ 20 – $30\times$ slower than bge.

5 Where each system LOST

The project contract: report losses, not just wins.

- A** **BM25**: lost everywhere on accuracy — lexical overlap $\neq$ intent. The floor, kept for reference.
- B** **Prompted llama3.2:3b** (deployed model): lost to both fine-tunes on every independent metric and is the **slowest on-device** system. Its only high number — agreement with the system's own past grades — is *self-reproduction*, not accuracy.
- C** **bge cross-encoder**: **LOST out-of-domain** — WANDS 0.299 vs the LoRA's 0.486. Also a train/test gap (dev QWK peaked 0.471 vs held-out ESCI 0.360 — the honest number is the held-out one).
- D** **LoRA**: lost on speed/memory (~ 20 – $30\times$ slower than bge) and only *typed* bge in-domain on ESCI (0.353 vs 0.360). Its win is generalization.
- R2** **Qwen2.5-7B (GPU)**:
 - Losses to the LoRA out-of-domain: WANDS 0.354 vs 0.486** — that number stands.
 - Loses on **exact accuracy** to both fine-tunes (0.343/0.332 vs bge 0.517/0.473, LoRA 0.483/0.603). Its QWK is respectable only because misses are ordinarily *close* (± 1 acc 0.79/0.87).
 - Under-predicts grade 3** (predicts 3 for only 479/2,997 ESCI, 148/3,000 WANDS): agrees on *rank* far better than on *label*.
 - Needs a ≥ 24 GB datacenter GPU + ~ 104 s load — bge still wins the efficiency tradeoff.

6 Limitations & honest caveats

- L1 — Absolute QWK is modest.** Everything sits at ~ 0.3 – 0.5 : real improvements over baseline, **not a solved problem**.
- L2 — Compare systems WITHIN a dataset only.** WANDS is a coarser 3-class space (0/2/3, no grade 1), so a WANDS number and an ESCI number are not on the same ruler.
- L3 — Independent labels = accuracy; system history = reproduction.** 92.3% of past grades came from the same local model — excluded from every accuracy claim.
- L4 — This is an ON-DEVICE contest.** The system's shipped default is cloud-first (a hosted Gemini model), not benchmarked and likely to beat every laptop-bound system. The claim is “no API, runs on this machine, and beats the local judge” — not “beats frontier cloud models.”
- L5 — Round-2 family confound (open).** Qwen2.5-7B (open) stood in for Llama-3.1-8B (gated), so the $+0.073$ / $+0.110$ mixes family with size. A clean same-family 3B $\rightarrow$ 8B ablation is still open.
- L6 — Latency methods differ across systems;** even normalized, the ordering (bge $\ll$ LoRA $\ll$ prompted) holds.
- L7 — 14B/70B not run** — a deliberate choice: the 3B $\rightarrow$ 7B trend already showed diminishing returns inside the band.

7 Conclusion & impact

For an on-device search-relevance judge, **fine-tuning a small specialist (bge cross-encoder) is the best accuracy-per-millisecond** — better than the deployed model, ~ 1 – 2 orders of magnitude faster, $\sim 5\times$ lighter.

A bigger *prompted* model on a datacenter GPU buys **in-domain parity with no training**, but not a breakthrough: the **~ 0.3 – 0.5 QWK ceiling is a property of the task**, and the best out-of-domain result comes from a **3B LoRA on a laptop**, not from more GPU.

Practical takeaway: reach for the GPU to train a small specialist or to skip training via a mid-size prompted model — but don't expect raw model size to break the quality ceiling on pointwise relevance.

Reproducibility

All systems share one evaluation framework (`scripts/run_arm.mjs` , `src/evaluate.mjs`); label mapping imports the production RELEVANCE_SCALE; datasets are public ESCI and WANDS. Round-2 held prompt/parser/QWK byte-identical to Round-1 System B and ran on NRP/Nautilus via vLLM. Journals 00 — 09 document every decision, including the discarded first System-B run and the provider-mix correction. Repo: `experiments/relevancy-replacement/`.